

APPLICATION NOTE

Linux 通过配置文件调节发射功率

**ATBM6062C****1x1 802.11b/g/n
Wi-Fi 芯片**

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1 按频段配置功率

1.1 按照不同频段配置功率

命令: iwpriv wlan0 common setttxpower_byfile

参数: 1: 使用配置文件中的 deltagain; 0: 使用芯片 efuse 中的 deltagain;

注意: 必须在发包之后使用 setttxpower_byfile 命令才能生效

(1) 说明

Atbm wifi 驱动目前支持加载驱动时读取配置文件来调节当前发射功率, 功率调节分别满足高中低信道。配置文件的存放位置及文件名可以通过 atbm_wifi_driver\hal_apollo 目录下的 Makefile 来进行配置:

```
16: #####
17: #ccflags-y += -DCONFIG_ATBM_APOLLO_5GHZ_SUPPORT
18: #ccflags-y += -DCONFIG_ATBM_APOLLO_WAPI_SUPPORT
19: #ccflags-y += -I$(shell pwd)/../include/ \
20:     -include $(shell pwd)/../include/linux/compat-2.6.h \
21:     -DCOMPAT_STATIC
22: ##指定配置文件路径, 当配置文件存在时使用配置文件中的dcxo值不使用efuse中的dcxo, 配置文件中的delta_gain为修i
23: ATBM_CONFIG_FILE="/tmp/atbm_txpwr_dcxo_cfg.txt"
24: ccflags-y += -DCONFIG_TXPOWER_DCXO_VALUE="\$(ATBM_CONFIG_FILE)\\"
25: # use the deltagain value in file :"\$(ATBM_CONFIG_FILE)",when file exist
26: ATBM_RATE_POWER_CONFIG_FILE="/tmp/set_rate_power.txt"
27: ccflags-y += -DCONFIG_RATE_TXPOWER="\$(ATBM_RATE_POWER_CONFIG_FILE)\\"
28:
29: ifneq ($(TF1NAMEF).)
```

文件内容格式如下:

格式要求: **必须按照如下格式配置, 打乱顺序会导致配置错位。**

delta_gain1:1

delta_gain2:2

delta_gain3:3

dcxo:59

```
1 delta_gain1:1
2 delta_gain2:2
3 delta_gain3:3
4 dcxo:59
```

数据中三个 delta gain 分别是调节低中高三段信道的功率因子, delta_gain1:1 调节信道 1~信道 4; delta_gain2:调节信道 5~信道 9; delta_gain3:调节信道 10~信道 14; dcxo:调节频偏。这三个 delta gain 不区分模式, 则在发包时就会使用当前配置的值在 efuse 的基础上补偿并立即生效。

第一行数据有效范围 (即调节范围 [-3.75dB~3.75dB]):

名称	有效范围	调节范围	物理意义
----	------	------	------

delta_gain1	-15~15	信道 1~信道 4	(delta_gain1/4)dB
delta_gain2	-15~15	信道 5~信道 9	(delta_gain2/4)dB
delta_gain3	-15~15	信道 10~信道 14	(delta_gain3/4)dB
dcxo	0~127	频偏	无

特别注意：因为是在原有 efuse delta_gain 的基础上做补偿，所以 efuse 里的 delta_gain 加上配置文件里的 deltagain 也必须要在【-3.75， 3.75】范围内。

(2) 配置方式

在芯片 efuse 原有 deltagain 的基础上做补偿

(3) 注意

数据都不会写到 efuse，该配置掉电就没了，不会保存到 ROM

PS:

驱动在加载的时候会默认去读取这个文件并根据配置文件内容生效。

运行过程中修改了配置文件内容，并想立即生效直接执行命令：

```
iwpriv wlan0 common settxpower_byfile
```

驱动会去刷新配置并立即生效。

2 按速率配置功率

2.1 修改不通速率的功率

命令：iwpriv wlan0 common set_rate_power,<useflag>

useflag:是否使用从配置文件中读取的数据，1：使用；0：使用默认发送功率

(1) 配置方式

配置文件的存放位置及文件名可以通过 atbm_wifi_driver\hal_apollo 目录下的 Makefile 来进行配置：

```

16: ##### 默认使用模式 #####
17: #ccflags-y += -DCONFIG_ATBM_APOLLO_5GHZ_SUPPORT
18: #ccflags-y += -DCONFIG_ATBM_APOLLO_WAPI_SUPPORT
19: #ccflags-y += -I$(shell pwd)/../include/ \
20:     -include $(shell pwd)/../include/linux/compat-2.6.h \
21:     -DCOMPAT_STATIC
22: ##指定配置文件路径，当配置文件存在时使用配置文件中的dcxo值不使用efuse中的dcxo，配置文件中的delta_gain为修正值
23: ATBM_CONFIG_FILE="/tmp/atbm_txpwer_dcxo_cfg.txt"
24: ccflags-y += -DCONFIG_TXPOWER_DCXO_VALUE="\$(ATBM_CONFIG_FILE)\\"
25: use the delta gain value in file "\$(ATBM_CONFIG_FILE)" when file exist
26: ATBM_RATE_POWER_CONFIG_FILE="/tmp/set_rate_power.txt"
27: ccflags-y += -DCONFIG_RATE_TXPOWER="\$(ATBM_RATE_POWER_CONFIG_FILE)\\"
28:

```

配置文件格式:

```

target_or_delta=1
b_1M_2M=210
b_5_5M_11M=210
g_6M_n_6_5M=210
g_9M=200
g_12M_n_13M=190
g_18M_n_19_5M=190
g_24M_n_26M=190
g_36M_n_39M=190
g_48M_n_52M=190
g_54M_n_58_5M=180
n_65M=180
HE-SU_MCS0=210
HE-SU_MCS1=190
HE-SU_MCS2=190
HE-SU_MCS3=190
HE-SU_MCS4=190
HE-SU_MCS5=190
HE-SU_MCS6=180
HE-SU_MCS7=180
HE-SU_MCS8=170
HE-SU_MCS9=170
HE-SU_MCS10=160
HE-SU_MCS11=160
g_6M_n_6_5M_40M=200
g_9M_40M=190
g_12M_n_13M_40M=190
g_18M_n_19_5M_40M=190
g_24M_n_26M_40M=190
g_36M_n_39M_40M=190
g_48M_n_52M_40M=190
g_54M_n_58_5M_40M=180
n_65M_40M=170
HE-SU_MCS0_40M=200
HE-SU_MCS1_40M=190
HE-SU_MCS2_40M=190
HE-SU_MCS3_40M=190

```

```
HE-SU_MCS4_40M=190
HE-SU_MCS5_40M=190
HE-SU_MCS6_40M=180
HE-SU_MCS7_40M=170
HE-SU_MCS8_40M=170
HE-SU_MCS9_40M=170
HE-SU_MCS10_40M=160
HE-SU_MCS11_40M=160
```

如果不使用 HT40 模式直接将这部分的功率配置直接写 0 即可。

target_or_delta 只能配置为 1，调试指定绝对发送功率

示例：

target_or_delta=1:

数据配置后仍应该使得发送功率在 20dbm 以内，配置的数据是 10*dB。

如果设置 b_1M_2M=180，则 1/2M 速率发射功率为 18dBm。

如果设置 g_6M_n_6_5M =140，则 6M/6.5M 速率发射功率为 14dBm。

(2) 注意

- 1) 驱动在加载的时候默认去读取这个文件并根据配置文件内容生效。

运行过程中修改了配置文件内容，并想立即生效直接执行命令：

```
iwpriv wlan0 common set_rate_power,1
```

驱动会去刷新配置并立即生效。

使用命令将设置好的配置复位：

```
iwpriv wlan0 common set_rate_power,0
```

3 获取当前的配置信息

3.1 通过 iwpriv 方式

命令格式为：

```
iwpriv wlan0 common get_cfg_txpower
```

```

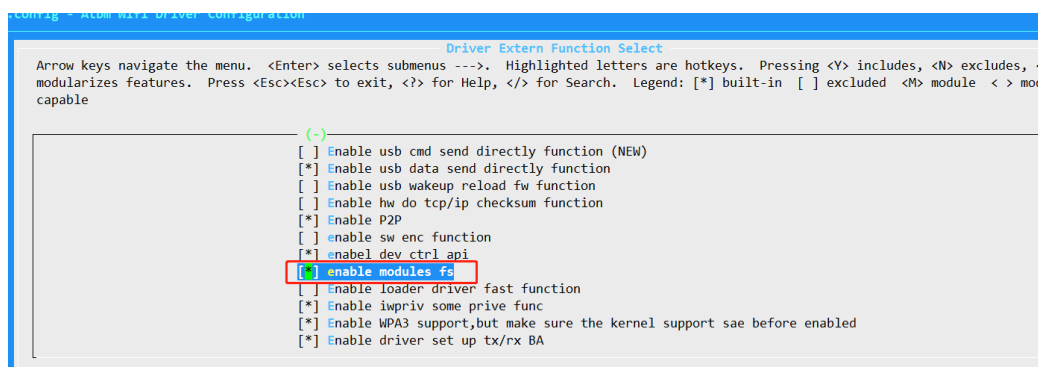
[atom_fog]:HE-SU_MCS11_40M=0
wlan0 common:b_1M_2M=0
b_5M_11M=0
g_6M_n_6_5M=0
g_9M=0
g_12M_n_13M=0
g_18M_n_19_5M=0
g_24M_n_26M=0
g_36M_n_39M=0
g_48M_n_52M=0
g_54M_n_58_5M=0
n_65M=0
HE-SU_MCS0=0
HE-SU_MCS1=0
HE-SU_MCS2=0
HE-SU_MCS3=0
HE-SU_MCS4=0
HE-SU_MCS5=0
HE-SU_MCS6=0
HE-SU_MCS7=0
HE-SU_MCS8=0
HE-SU_MCS9=0
HE-SU_MCS10=0
HE-SU_MCS11=0
g_6M_n_6_5M_40M=0
g_9M_40M=0
g_12M_n_13M_40M=0
g_18M_n_19_5M_40M=0
g_24M_n_26M_40M=0
g_36M_n_39M_40M=0
g_48M_n_52M_40M=0
g_54M_n_58_5M_40M=0
n_65M_40M=0
HE-SU_MCS0_40M=0
HE-SU_MCS1_40M=0
HE-SU_MCS2_40M=0
HE-SU_MCS3_40M=0
HE-SU_MCS4_40M=0
HE-SU_MCS5_40M=0
HE-SU_MCS6_40M=0
HE-SU_MCS7_40M=0
HE-SU_MCS8_40M=0
HE-SU_MCS9_40M=0
HE-SU_MCS10_40M=0
HE-SU_MCS11_40M=0

delta_gain1:27 delta_gain2:27 delta_gain3:27 dcxo:50

```

3.2 通过 module_fs 方式获取

需要 ATBM_WIFI 驱动打开 MODULES_FS 宏



```
cat /sys/module/<MODULES_NAME>/atbmfs/ atbm_get_cfg_power
```

当前距离驱动的 MODULES_NAME 为:

```
/sys/module/ATBM606x_wifi_usb/atbmfs #
/sys/module/ATBM606x_wifi_usb/atbmfs # lsmod
ATBM606x_wifi_usb 626103 0 - Live 0xbfc55000 (o)
cfg80211 285288 1 ATBM606x_wifi_usb, Live 0xbfb74000 (o)
nls_utf8 1380 0 - Live 0xbfb70000
exfat 64522 0 - Live 0xbfb59000
drv_ms_cus_gc2053_MIPI 6865 0 - Live 0xbfb53000 (PO)
mi_shadow 34439 0 - Live 0xbfb45000 (PO)
mi_venc 165330 0 - Live 0xbfb12000 (PO)
mi_vif 61952 0 - Live 0xbfafc000 (PO)
mi_sensor 22016 0 - Live 0xbfaf2000 (PO)
mi_vpe 174963 0 - Live 0xbfabd000 (PO)
mi_div 48528 0 - Live 0xbfab5000 (PO)
```

所以整体的执行命令为:

```
cat /sys/module/ATBM606x_wifi_usb/atbmfs/atbm_get_cfg_power
```

```
/tmp # cat /sys/module/ATBM606x_wifi_usb/atbmfs/atbm_get_cfg_power
b_1M_2M=0
b_5_5M_11M=0
g_6M_n_6_5M=0
g_9M=0
g_12M_n_13M=0
g_18M_n_19_5M=0
g_24M_n_26M=0
g_36M_n_39M=0
g_48M_n_52M=0
g_54M_n_58_5M=0
n_65M=0
HE-SU_MCS0=0
HE-SU_MCS1=0
HE-SU_MCS2=0
HE-SU_MCS3=0
HE-SU_MCS4=0
HE-SU_MCS5=0
HE-SU_MCS6=0
HE-SU_MCS7=0
HE-SU_MCS8=0
HE-SU_MCS9=0
HE-SU_MCS10=0
HE-SU_MCS11=0
g_6M_n_6_5M_40M=0
g_9M_40M=0
g_12M_n_13M_40M=0
g_18M_n_19_5M_40M=0
g_24M_n_26M_40M=0
g_36M_n_39M_40M=0
g_48M_n_52M_40M=0
g_54M_n_58_5M_40M=0
n_65M_40M=0
HE-SU_MCS0_40M=0
HE-SU_MCS1_40M=0
HE-SU_MCS2_40M=0
HE-SU_MCS3_40M=0
HE-SU_MCS4_40M=0
HE-SU_MCS5_40M=0
HE-SU_MCS6_40M=0
HE-SU_MCS7_40M=0
HE-SU_MCS8_40M=0
HE-SU_MCS9_40M=0
HE-SU_MCS10_40M=0
HE-SU_MCS11_40M=0

delta_gain1:27 delta_gain2:27 delta_gain3:27 dcxo:50
/tmp #
```




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